# Data Models

## Document Modelling

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8 February 2026

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# Building On: Topics 2–5

Topics 2–5 produced a complete relational toolkit:

- **Topic 2:** ERD → structurally correct schema (PKs, FKs, cardinality constraints)
- **Topic 3:** BCNF normalisation eliminated redundancy; justified downshifts documented
- **Topic 4:** FWGHOS pipeline queries the schema correctly
- **Topic 5:** Constraints, transactions, indexes, and access control harden the schema

The toolkit is complete — for **transactional** workloads. But not every workload in a production system is transactional.

# The Super-App: Two Workloads, One Platform

```
SUPER-APP PLATFORM

TRANSACTIONAL CORE

Ride matching
Payment processing
Settlement
Promo enforcement

▶ ACID txns
▶ FK integrity
▶ Normalised

Relational model
(Topics 2–5) ✓

PERSONALISATION LAYER

Passenger profiles
Driver profiles
Merchant dashboards
Preferences & rewards

▶ Aggregate reads
▶ Structural variation
▶ Nested JSON delivery

??? (this topic)
```

The left side is solved. The right side is where the relational model starts to resist.

# Failure 1: The Schema Resists Structural Variation

Three user roles — passenger, driver, merchant — each with fundamentally different profile data. The developer considers three relational options:

| Option | Design | Problem |
|---|---|---|
| **Wide table** | All columns in one `UserProfile` (40+ cols) | Sparse rows; schema no longer communicates what a "driver" looks like |
| **EAV** | `UserAttribute(user_id, attr_name, attr_value)` | Type safety lost — `licenseExpiry` (date) stored as string |
| **Subtype tables** | `PassengerProfile`, `DriverProfile`, `MerchantProfile` | Rigid — new role means new table, new joins, new app logic |

All three are valid relational designs. None handles what comes next.

# The Combinatorial Explosion

The product team adds tier-specific fields *within* each role:

- Premium passengers: `loyaltyTier`, `loungeAccess`
- GrabCar drivers ≠ GrabFood riders: `thermalBagCertified`, `bikeType`
- Subscription merchants: `featuredPlacement`, `analyticsAccess`

Role × tier × feature flags → each combination needs nullable columns or subtype rows. Every product sprint ships a new variant; every variant ships a migration.

**Result:** 14 schema migrations in 6 months for profile changes alone. Two cause 30-minute read-only periods during peak hours. ~3–4 engineering-days per sprint on migration logistics.

The relational model treats structural variation as a **schema change**. The application treats it as **data**.

# Failure 2: Storage Shape ≠ Delivery Shape

The mobile app needs a driver's complete profile on login — one nested JSON response:

```
Relational storage
(4 tables, 3+ JOINs)

┌─ UserProfile ──────────────┐
│ user_id │ display_name     │
│ phone   │ email            │
└────────────────────────────┘
              │ FK
┌─ DriverProfile ────────────┐
│ user_id │ vehicle_make     │
│ vehicle_model │ plate      │
│ license_number             │
│ license_expiry             │
└────────────────────────────┘
              │ FK
┌─ PreferredZones ───────────┐
│ user_id │ zone_name        │
├────────────────────────────┤
│ d-5582  │ central          │
│ d-5582  │ east             │
└────────────────────────────┘
              │ FK
┌─ TripHistory ──────────────┐
│ user_id │ trip_id          │
│ route   │ fare │ date      │
└────────────────────────────┘
```

```
Profile-load query:
SELECT ...
FROM UserProfile
INNER JOIN DriverProfile
INNER JOIN PreferredZones
INNER JOIN TripHistory
WHERE user_id = ...
→ flat rows → app re-nests JSON
```

```
Document storage
(1 document, 1 read)

┌─ users collection ─────────────────────────┐
│ {                                          │
│   "userId": "driver-5582",                 │
│   "displayName": "Ahmad ... ",             │
│   "phone": "+6598765432",                  │
│   "vehicle": { ... },                      │
│   "license": { ... },                      │
│   "preferences": { ... },                  │
│   "earnings": { ... },                     │
│   "subscriptionSnapshot": ...              │
│ }                                          │
└────────────────────────────────────────────┘
```

```
Profile-load query:
db.users.findOne({
 "userId": "driver-5582"
})
→ returns complete nested JSON
```

Three user types × three different JOIN chains × three re-nesting implementations — all bridging the same shape mismatch.

# The Re-Nesting Tax

Each new nested structure (e.g., `incentiveProgress` inside earnings) requires updating re-nesting code for all three user types:

- **2–4 hours** of re-nesting code per user type per structural addition
- **3 user types × monthly additions → 8–12 engineering-hours/month**
- Produces **no user-visible value** — exists only to bridge the mismatch

The schema is correct. The queries are correct. The data is consistent. But the transformation layer between storage and delivery exists only because the **storage shape doesn't match the delivery shape**.

# The Root Cause

The relational model is **not wrong** for this platform. The transactional core needs exactly the guarantees from Topics 2–5.

The root cause: the platform has a **second workload** — the personalisation layer — with two characteristics the relational model resists:

1. **Structural variation is routine** — new roles, tiers, features every sprint. The relational model treats this as schema change; the application treats it as data.
2. **The dominant read is an aggregate** — the app reads whole profiles as one unit. The normalised schema forces disassembly on write and reassembly on read.

This parallels Topic 3: normalisation was correct for OLTP but wrong for OLAP reporting — the answer was a second schema (star schema) shaped for the workload. Same principle, larger scale.

# The Pattern

> Therefore, the engineer must **choose a storage model shaped for the personalisation workload** — not discover the mismatch after 14 migrations and months of re-nesting code.

The transactional core stays relational. The personalisation layer needs a storage shape that matches the delivery shape — different data models for different jobs.

What must this storage design guarantee?

# Three Requirements for Aggregate Storage

| # | Requirement | Traces to |
|---|---|---|
| **R1** | **Store data in the shape the application reads it** — one read returns the complete document; no joins, no re-nesting | Failure 2: storage-delivery shape mismatch |
| **R2** | **Accommodate structural variation without schema migration** — new roles, tiers, features are data, not schema changes | Failure 1: 14 migrations in 6 months |
| **R3** | **Make the integrity trade-offs explicit** — state what relational guarantees are preserved, lost, and how gaps are managed | Topics 2–5: hard-won guarantees must not be silently abandoned |

R1 demands a new storage shape. R2 demands a new schema posture. R3 demands honesty about what is gained and what is lost.

# The Document Model

| | |
|---|---|
| **Unit of storage** | Document — a self-contained JSON object stored as a unit |
| **Connection mechanism** | Embedding (nesting) within documents; references (`_id`) across documents |
| **Schema enforcement** | Schema-on-read by default — "nothing rejects" on the enforcement spectrum |

A **document** is a self-contained JSON object. Documents are grouped into **collections** — the document analogue of a relational table. Unlike rows, documents in the same collection can have different fields.

Every document has a unique `_id` field — the same role as Topic 2's primary key.

# JSON as the Single Notation

In the relational world, the data language and query language are separate (DDL, DML, result sets). In the document model, everything is JSON:

| Role | Relational | Document Model |
|---|---|---|
| Define Structure | DDL (`CREATE TABLE`) | Implicit in the documents themselves |
| Store Data | Rows with fixed columns | JSON documents with flexible fields |
| Query / Filter | SQL `WHERE` clause | JSON filter: `{ "role": "driver" }` |
| Choose Output Fields | SQL `SELECT` list | JSON projection: `{ "displayName": 1 }` |
| Transform / Aggregate | SQL `GROUP BY` + functions | JSON pipeline: `[{ "$group": ... }]` |

The storage shape, the query shape, and the API response shape are all the same notation.

# CRUD Operations: MongoDB vs SQL

| Operation | SQL (Relational) | MongoDB (Document) |
|---|---|---|
| **Create** | `INSERT INTO users ( ... ) VALUES ( ... )` | `db.users.insertOne({ ... })` |
| **Read one** | `SELECT * FROM users WHERE user_id = 'p-1001'` | `db.users.findOne({ "userId": "p-1001" })` |
| **Read many** | `SELECT * FROM users WHERE role = 'driver'` | `db.users.find({ "role": "driver" })` |
| **Update** | `UPDATE users SET name = 'Sarah T' WHERE ...` | `db.users.updateOne({ ... }, { "$set": { "displayName": "Sarah T" } })` |
| **Delete** | `DELETE FROM users WHERE user_id = 'p-1001'` | `db.users.deleteOne({ "userId": "p-1001" })` |
| **Count** | `SELECT COUNT(*) FROM users` | `db.users.countDocuments({})` |

`$set` updates a scalar field; `$push` appends to an array. Both happen atomically within a single `updateOne`.

Return values confirm what happened:

- `insertOne` → `{ acknowledged: true, insertedId: ObjectId(" ... ") }`
- `updateOne` → `{ acknowledged: true, matchedCount: 1, modifiedCount: 1 }`

# Aggregate and Aggregate Root

> An **aggregate** is a cluster of related objects treated as a unit for reads and writes. The **aggregate root** is the top-level entity — everything nested inside belongs to that aggregate.

The aggregate boundary defines the document boundary. In the relational model, the unit of storage is the *row* and relationships are reconstructed via joins. In the document model, the unit of storage is the *aggregate* and relationships within it are represented by nesting.

Relationships that **cross** aggregate boundaries — a user profile referencing a shared subscription plan — require references.

# Embed vs Reference

| Strategy | Structure | Use when | Consequence |
|---|---|---|---|
| **Embed** | Nest child inside parent document | Retrieved together, updated together, bounded in size | One read, one atomic write. Risk: document growth if unbounded |
| **Reference** | Store child in separate collection; parent holds child's `_id` | Shared across many parents, updated independently, or unbounded growth | Requires second query or `$lookup`. No single-document atomicity across the boundary |

The decision is **workload-driven**, not structural. The same data might be embedded in one context (snapshot of plan terms at sign-up) and referenced in another (live plan definitions in the `plans` collection).

# Snapshot Fields

> A **snapshot field** is an immutable copy of data captured at a point in time — recording a contractual fact, not a copy of a mutable fact.

A driver's `planNameAtSignUp` and `commissionRateAtSignUp` record what terms the driver agreed to. The app displays "You signed up under GrabCar Premium at 20% commission" — this mustn't change when the company adjusts the rate to 18%.

This is Topic 3's shipping-address snapshot in document form: label the snapshot, identify the truth, never update the snapshot. Any unlabelled duplication is a bug.

# Bounded vs Unbounded Embedding

**Embed** when the array has a known upper bound:

- Saved places per passenger: ≤ 10
- Payment methods per user: ≤ 5
- Menu items per merchant: ≤ 100

**Reference** when the array could grow without limit:

- All ride history for a daily commuter over years
- All order history for a popular merchant across months

The bounded-growth rule: state the **upper bound**, the **business rule** that enforces it, and the **fallback** if exceeded. This is the document-model analogue of Topic 2's (min, max) cardinality discipline.

# Schema-on-Read vs Schema-on-Write

| | Schema-on-Write (Relational) | Schema-on-Read (Document) |
|---|---|---|
| Structure defined | Before insert (DDL) | By the documents themselves |
| Variation across rows/docs | Not allowed without ALTER TABLE | Natural — each document carries its own fields |
| Invalid data | Rejected by the database | Accepted unless validation rules added |
| Schema evolution | Requires migration | New fields appear in new documents |
| Enforcement cost | Paid once (at write) | Paid on every read (or added via validation) |

Schema-on-read makes Failure 1 disappear — adding a new role is just inserting documents with new keys. Cost: nothing prevents a misspelled `"preferedZones"` from being accepted silently.

# Enforcement Spectrum — Document Model Position

```
Nothing                    Validator                  Engine
rejects                    rejects                    rejects
◀──────────────────────────────────────────────────────────────────▶

  MongoDB                  MongoDB with              PostgreSQL
  (default)                $jsonSchema               (DDL + constraints)
     ▲                        ▲                          ▲
     |                        |                          |
  Schema-on-read           Opt-in                    Schema-on-write
  Any JSON accepted        validation rules          Every row checked
  App interprets           reject invalid docs       Engine rejects bad data
```

MongoDB can partially close the enforcement gap with `$jsonSchema` validation — but the default posture is permissive. Enforcement must be **chosen deliberately**.

# The Permissive Default

You insert `{ "displayName": "Test" }` into a MongoDB collection with **no validation rules** — no `userId`, no `role`, no type checking.

Then you insert a row into the relational `UserProfile` table without a `user_id` (defined as `NOT NULL PRIMARY KEY`).

> **Predict:** What happens in each case?

# The Enforcement Spectrum in Action

> MongoDB **accepts** the document — it stores whatever JSON you provide. PostgreSQL **rejects** the insert with a NOT NULL violation.

The document stored in MongoDB is structurally valid JSON but semantically meaningless — no `userId` means no way to look up this profile. The relational system prevents this at the schema level; the document system requires either application logic or an explicit `$jsonSchema` validation rule.

```
// Adding validation — closing the enforcement gap for
critical fields
db.createCollection("users_validated", {
  validator: { $jsonSchema: {
    bsonType: "object",
    required: ["userId", "role"],
    properties: {
      role: { enum: ["passenger", "driver", "merchant"]
}
    }
  }}
})
```

```
// Missing required fields:
// MongoServerError: Document failed validation
// details: { schemaRulesNotSatisfied: [
//   { operatorName: "required",
//     missingProperties: ["userId", "role"] }] }

// Invalid role value:
// MongoServerError: Document failed validation
// details: { propertiesNotSatisfied: [
//   { propertyName: "role",
//     details: [{ operatorName: "enum",
//                 reason: "value was not found in enum" }] }]
}
```

Cost: the corporate admin role (`"corporate_admin"`) would be rejected until the enum is updated — validation reintroduces some migration friction.

# Shape Alignment: The Driver Profile as a Document

The API response shape **is** the storage shape:

```
{
  "userId": "driver-5582",
  "displayName": "Ahmad Ibrahim",
  "phone": "+6598765432",
  "role": "driver",
  "driverType": "grabcar",
  "vehicle": {
    "make": "Toyota", "model": "Prius", "year": 2022,
    "plateNumber": "SBA1234X"
  },
  "license": { "licenseNumber": "S1234567A", "expiryDate": "2026-08-15" },
  "preferences": { "maxDistance": 25, "preferredZones": ["central", "east"] },
  "earnings": { "currentWeek": 1240.50,
    "incentiveProgress": { "ridesCompleted": 42, "target": 50, "bonus": 120.00 }
  },
  "subscriptionSnapshot": {
    "planNameAtSignUp": "GrabCar Premium",
    "commissionRateAtSignUp": 0.20, "signUpDate": "2024-01-15"
  }
}
```

`db.users.findOne({ "userId": "driver-5582" })` — one read, complete profile, no transformation.

# Design A (Embedded) vs Design B (Referenced)

```
Design A — Embedded                    Design B — Referenced

┌── users ──────────────────┐            ┌── users_b ──────┐
│  userId, role             │            │  userId, role   │
│                           │            └─────────────────┘
│  ┌ vehicle: {}  ┐         │                    │ userId
│  ├ license: {}  ┤         │          ┌─────────┴─────────┐
│  ├ prefs: {}    ┤         │          ▼         ▼         ▼
│  ├ earnings: {} ┤         │     vehicles   prefs    earnings
│  └ snapshot: {} ┘         │        _b       _b        _b
└───────────────────────────┘

findOne → 1 read                       findOne + 3× $lookup → 4+ reads
Returns complete profile               Application reassembles JSON
```

Design A aligns with the profile-load workload (R1). Design B recreates the relational pattern — JOIN chains and re-assembly — inside the document model.

# Three Roles, One Collection: Schema-on-Read

```
{ "userId": "passenger-1001", "role": "passenger",
  "ridePreferences": { "vehicleType": "JustGrab", "quietRide": false },
  "savedPlaces": [
    { "label": "Home", "address": "123 Clementi Ave 3" },
    { "label": "Office", "address": "1 Raffles Place" }
  ],
  "rewardPoints": 2840 }

{ "userId": "driver-5582", "role": "driver", "driverType": "grabcar",
  "vehicle": { "make": "Toyota", "model": "Prius", "plateNumber": "SBA1234X" },
  "license": { "licenseNumber": "S1234567A", "expiryDate": "2026-08-15" },
  "preferences": { "maxDistance": 25, "preferredZones": ["central", "east"] } }

{ "userId": "merchant-0891", "role": "merchant",
  "businessName": "Ah Huat Chicken Rice",
  "menu": [{ "itemName": "Roasted Chicken Rice", "price": 5.50, "category": "mains",
    "customisations": [{ "name": "Extra rice", "price": 0.50 }] }] }
```

No NULLs, no EAV, no subtype tables. Adding a fourth role is just inserting documents with new keys (R2).

# Atomicity: Relational vs Document

A passenger opens the app, adds a new saved place, and earns 50 reward points simultaneously.

- In the **relational model** (Topic 5), this update touches **two tables** (`SavedPlaces` + `UserProfile`)
- In the **document model**, both fields live inside the **same document**

> **Predict:** What construct does the relational model need to make both changes atomic? What guarantees atomicity in the document model without that construct?

# Aggregate Boundary = Atomicity Boundary

> The relational model needs a **transaction** (`BEGIN`/`COMMIT` from Topic\ 5). The document model's single `updateOne` is **atomic by default** — both fields live in the same aggregate.

```
db.users.updateOne(
  { "userId": "passenger-1001" },
  { "$set": { "rewardPoints": 2890 },
    "$push": { "savedPlaces": { "label": "Gym", "address": "10 Bayfront Ave" } } }
)
// Both changes happen atomically — no BEGIN/COMMIT needed
```

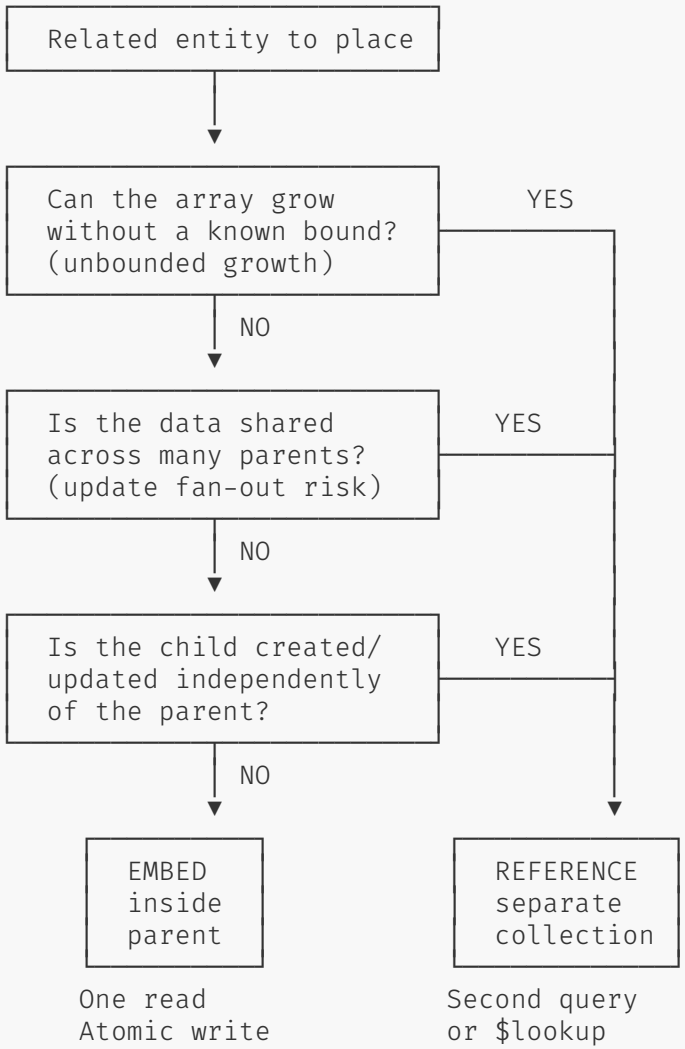
**Read locality:** one disk read retrieves parent + all children. **Single-document atomicity:** the aggregate boundary *is* the atomicity boundary. These are the two guarantees embedding provides.

# When Embedding Breaks: Three Conditions

| Condition | Test | Risk if violated |
|---|---|---|
| **Unbounded growth** | Can the array grow without a known bound? | Hits 16 MB BSON limit; read performance degrades before that |
| **Shared data** | Is the child shared across many parents? | **Update fan-out** — one logical change becomes thousands of physical writes |
| **Independent lifecycle** | Is the child created/updated independently of the parent? | Couples write paths — every child update rewrites the parent |

If **any** condition is true → **reference** instead of embedding. This is the **embed-or-reference test** — a design-time decision, not a runtime check.

# Embed-or-Reference Flowchart




Apply this to every related entity during document design.

# Ride History: Embed or Reference?

A driver's ride history grows by 5–10 entries per day. After one year: ~2,000–3,500 entries, each with route coordinates, fare breakdown, and timestamps.

The profile-load only needs the **5 most recent trips**.

> **Predict:** Should ride history be embedded or referenced? Which condition of the embed-or-reference test applies?

# Unbounded Growth → Reference

> **Referenced.** Unbounded growth — no business-rule upper bound. A 3-year driver accumulates 6,000–10,000 entries. Every profile-load would pull thousands of unnecessary entries.

Ride history belongs in a separate `rideHistory` collection, referenced by `userId`. The profile can embed `recentTrips` (last 5 — bounded) while the full history is referenced.

Costs accepted: the full-history query requires a second read or `$lookup`. The profile-load avoids the unbounded penalty.

# Snapshot Discipline: Truth Map

```
TRUTH MAP — Subscription plan data

  plans collection (EDITABLE TRUTH)        users collection (IMMUTABLE SNAPSHOT)

   planId: "plan-premium"                    userId: "driver-5582"
   name: "GrabCar Premium"                   ...
   commissionRate: 0.18  ◄                   subscriptionSnapshot: {
   tier: "premium"                             planNameAtSignUp:
   monthlyPrice: 29.99                           "GrabCar Premium"
                                               commissionRateAtSignUp: 0.20
                                               signUpDate: "2024-01-15"
                                             }

              Rate changed
              from 0.20
              to 0.18

                   1 write                     0 writes  ▲
                                               Snapshot is immutable —
                                         X ——— records terms at sign-up
```

Without snapshots, changing the commission rate is 5,000 writes (one per driver). With snapshots: **1 write**. This is Topic 3's truth-drift anomaly in document form — the fix is the same: label the copy, identify the truth, define the update rule.

# Seed Data: Users and Plans

The examples and queries that follow use this seed data (from the book's Tables 6.5–6.6).

**7 users** in the `users` collection:

| displayName | role | subscriptionSnapshot.planNameAtSignUp |
|---|---|---|
| Sarah Tan | passenger (standard) | — |
| Wei Ming Lim | passenger (premium) | GrabCar Premium |
| Rani Devi | passenger (standard) | — |
| Ahmad Ibrahim | driver (grabcar) | GrabCar Premium |
| Priya Nair | driver (grabfood) | GrabFood Partner |
| Ah Huat | merchant | GrabFood Partner |
| Jia Yi | merchant | GrabFood Standard |

**4 plans** in the `plans` collection:

| name | tier | monthlyPrice | commissionRate |
|---|---|---|---|
| GrabCar Premium | premium | SGD 29.99 | 0.20 |
| GrabCar Standard | standard | SGD 0.00 | 0.25 |
| GrabFood Partner | partner | SGD 0.00 | 0.30 |
| GrabFood Standard | standard | SGD 0.00 | 0.35 |

# Seed Data: Menu Items

**9 menu items** across both merchants (embedded in `menu[]`):

| Merchant | itemName | price | category |
|---|---|---|---|
| Ah Huat Chicken Rice | Roasted Chicken Rice | SGD 5.50 | mains |
| Ah Huat Chicken Rice | Steamed Chicken Rice | SGD 5.00 | mains |
| Ah Huat Chicken Rice | Char Siew Rice | SGD 6.00 | mains |
| Ah Huat Chicken Rice | Iced Barley | SGD 1.50 | drinks |
| Ah Huat Chicken Rice | Hot Tea | SGD 1.20 | drinks |
| Tiger Sugar Toa Payoh | Brown Sugar Boba | SGD 6.90 | signature |
| Tiger Sugar Toa Payoh | Tiger Black Sugar | SGD 5.90 | signature |
| Tiger Sugar Toa Payoh | Oolong Tea Latte | SGD 5.50 | tea |
| Tiger Sugar Toa Payoh | Matcha Latte | SGD 6.50 | tea |

After `$unwind` on `menu`, these 9 items become 9 separate documents — one per row above.

# Querying Nested Arrays: Two Queries, Same Data

"Find passengers where a single saved place has frequency 'daily' AND was last used before 2025-01-01."

```
// Query A — dot notation (conditions evaluated independently)
db.users.find({
  "savedPlaces.frequency": "daily",
  "savedPlaces.lastUsed": { "$lt": "2025-01-01" }
})

// Query B — $elemMatch (conditions bound to same element)
db.users.find({
  "savedPlaces": { "$elemMatch": {
    "frequency": "daily",
    "lastUsed": { "$lt": "2025-01-01" }
  }}
})
```

| Passenger | Saved places |
|---|---|
| Sarah Tan | Home (daily, 2025-01-20), Office (weekday, 2024-12-10) |
| Wei Ming Lim | Home (daily, 2025-01-18), Gym (weekly, 2025-01-15), Parents (monthly, 2024-12-25) |
| Rani Devi | (none) |

The seed data also includes `frequency` and `lastUsed` metadata on saved places (omitted from earlier slides for brevity).

> **Predict:** Does Query A match Sarah's document? Does Query B? Why might they differ?

# `$elemMatch`: Why the Queries Differ

Query A **matches** — "daily" comes from Home, "before 2025" comes from Office. Different elements satisfy different conditions. Query B **does not match** — no single element satisfies both.

```
Element 0: { label: "Home",   frequency: "daily",   lastUsed: "2025-01-20" }
Element 1: { label: "Office", frequency: "weekday", lastUsed: "2024-12-10" }

Query A (dot notation):        Query B ($elemMatch):
"daily" ← Element 0    √      Element 0: daily √, but lastUsed ✗ (too recent)
"< 2025" ← Element 1   √      Element 1: lastUsed √, but frequency ✗ (weekday)
Both satisfied → MATCH         No single element satisfies both → NO MATCH
```

| Passenger | Has "daily"? | Has lastUsed < 2025? | Same element? | Query A | Query B |
|---|---|---|---|---|---|
| Sarah Tan | Home ✓ | Office ✓ | No | **Match** | **No match** |
| Wei Ming | Home ✓ | Parents ✓ | No | **Match** | **No match** |
| Rani Devi | ✗ | ✗ | — | No match | No match |

In the relational model, `WHERE` naturally binds conditions to the same row. Embedding collapses rows into arrays — `$elemMatch` recovers per-element filtering. A cost of embedding.

# Regex Queries and Index Interaction

MongoDB supports regex filters on string fields using `/pattern/` short notation (equivalent to the explicit `{ "$regex": "pattern" }` form). The interaction with indexes follows Chapter 5's selectivity principle:

| Short Form | Explicit Form | Anchored? | Case-Sensitive? | Index Usage | SQL Equivalent |
|---|---|---|---|---|---|
| `/^\+659/` | `{ "$regex": "^\\+659" }` | ✓ prefix | ✓ | IXSCAN (B-Tree Seek) | `WHERE phone LIKE '+659%'` |
| `/^Ahmad/` | `{ "$regex": "^Ahmad" }` | ✓ prefix | ✓ | IXSCAN (B-Tree Seek) | `WHERE name LIKE 'Ahmad%'` |
| `/chicken rice/i` | `{ "$regex": "chicken rice", "$options": "i" }` | ✗ | ✗ | COLLSCAN (Full Scan) | `WHERE LOWER(name) LIKE '%chicken%'` |
| `/^Wei/i` | `{ "$regex": "^Wei", "$options": "i" }` | ✓ prefix | ✗ | COLLSCAN (Full Scan) | `WHERE LOWER(name) LIKE 'wei%'` |

Both forms are equivalent — `/^Ahmad/` and `{ "$regex": "^Ahmad" }` produce the same filter. Anchored, case-sensitive regexes use the index. The `"i"` flag forces case-insensitive matching — the engine can't narrow to one case range, even with a prefix anchor.

# The Aggregation Pipeline

The document equivalent of Topic 4's FWGHOS. Key differences:

- **Stage order is the author's choice** — FWGHOS has fixed logical order; the pipeline executes in the order you write it
- **`$unwind` has no SQL equivalent** — it deconstructs embedded arrays so `$group` can treat items from different documents as peers

```
[
  { "$match": { "role": "merchant" } },
  { "$unwind": "$menu" },
  { "$group": {
      "_id": "$menu.category",
      "itemCount": { "$sum": 1 },
      "avgPrice": { "$avg": "$menu.price" }
  }},
  { "$sort": { "itemCount": -1 } },
  { "$limit": 5 }
]
```

**Predict:** Using the seed data (2 merchants: Ah Huat with 5 menu items, Tiger Sugar with 4) — how many documents after `$match`? After `$unwind`? How many groups?

# Pipeline Trace: Stage by Stage

```
Stage 1: $match { "role": "merchant" }
  7 documents → 2 documents (Ah Huat + Tiger Sugar)

Stage 2: $unwind "$menu"
  2 documents → 9 documents (one per menu item):
  Ah Huat  → Roasted Chicken Rice  (mains, SGD 5.50)
           → Steamed Chicken Rice  (mains, SGD 5.00)
           → Char Siew Rice        (mains, SGD 6.00)
           → Iced Barley           (drinks, SGD 1.50)
           → Hot Tea               (drinks, SGD 1.20)
  Tiger S  → Brown Sugar Boba      (signature, SGD 6.90)
           → Tiger Black Sugar     (signature, SGD 5.90)
           → Oolong Tea Latte      (tea, SGD 5.50)
           → Matcha Latte          (tea, SGD 6.50)

Stage 3: $group by "$menu.category"
  9 documents → 4 groups:
  mains     → 3 items (5.50+5.00+6.00)/3 = SGD 5.50 avg
  drinks    → 2 items (1.50+1.20)/2      = SGD 1.35 avg
  signature → 2 items (6.90+5.90)/2      = SGD 6.40 avg
  tea       → 2 items (5.50+6.50)/2      = SGD 6.00 avg

Stage 4: $sort { "itemCount": -1 }
  mains (3) → drinks (2) → signature (2) → tea (2)

Stage 5: $limit 5
  All 4 pass (fewer than limit)
```

> `$match` → 2. After `$unwind` → 9. `$group` → 4. Like Topic 4's FWGHOS trace, but for the aggregation pipeline — and `$unwind` is the stage that has no SQL equivalent.

# `$lookup`: Crossing the Aggregate Boundary

> `$lookup` is the document model's equivalent of Topic 4's LEFT JOIN — preserving all "left" documents and attaching matches from the "right" collection.

```
db.users.aggregate([
  { "$match": { "role": "merchant" } },
  { "$lookup": {
      "from": "plans",
      "localField": "subscriptionSnapshot.planNameAtSignUp",
      "foreignField": "name",
      "as": "planInfo"
  }},
  { "$unwind": "$planInfo" },
  { "$unwind": "$menu" },
  { "$group": { "_id": "$planInfo.tier",
      "avgPrice": { "$avg": "$menu.price" }, "itemCount": { "$sum": 1 } }},
  { "$sort": { "avgPrice": -1 } }
])
```

6 stages here vs 5 for the within-aggregate query. The extra `$lookup` + `$unwind` is the cost of crossing the aggregate boundary. No free lunch: embedding is the default, referencing is the escape hatch — and the escape hatch is costlier than the relational equivalent.

# `$lookup` Pipeline Result

The pipeline resolves each merchant's `subscriptionSnapshot.planNameAtSignUp` to the matching plan, then groups by tier:

| _id (tier) | avgPrice | itemCount | Merchants in tier |
|---|---|---|---|
| standard | SGD 6.20 | 4 | Tiger Sugar (GrabFood Standard) |
| partner | SGD 3.84 | 5 | Ah Huat (GrabFood Partner) |

Tiger Sugar's bubble tea (avg SGD 6.20 across 4 items) is pricier per item than Ah Huat's hawker fare (avg SGD 3.84 across 5 items). The `$lookup` resolved plan names to tiers; `$unwind` deconstructed both the `$lookup` result and the `menu` array before `$group` aggregated by tier.

# The Trade-Off Ledger

| Guarantee | Relational (Ch 2–5) | Document Model | Status |
|---|---|---|---|
| Row/doc uniqueness | PRIMARY KEY | `_id` field | Preserved |
| Referential integrity | FOREIGN KEY | Not enforced — app must verify | Lost; app manages |
| Cross-entity constraints | CHECK, UNIQUE across tables | Within single doc only | Partially lost |
| Multi-step atomicity | Transactions (BEGIN/COMMIT) | Single-doc atomic; multi-doc costlier | Preserved within aggregate |
| No redundancy | BCNF by design | Deliberate snapshots; uncontrolled duplication is still a bug | Shifted; snapshot discipline |
| Ad-hoc joins | JOIN on any FK path | `$lookup` — functional but costlier | Available but costlier |
| Schema enforcement | DDL + constraints | Schema-on-read; opt-in `$jsonSchema` | Shifted; opt-in |

For each "lost" or "shifted" guarantee, the design must name a management strategy. No strategy → incomplete design (R3).

This ledger applies to the *personalisation layer*. The transactional core retains the full relational guarantee set. A production super-app uses both: relational for transactions, documents for personalisation — **polyglot persistence**.

# Callback: The Enforcement Spectrum (Now Concrete)

Earlier we asked where the document model sits on the enforcement spectrum. Now we can answer with specifics:

```
Nothing                          Validator                        Engine
rejects                          rejects                          rejects
◀──────────────────────────────────────────────────────────────────────────▶

  MongoDB                      MongoDB with                   PostgreSQL
  (default)                    $jsonSchema                    (DDL + constraints)
     ▲                              ▲                              ▲
     |                              |                              |
  { "displayName":             Missing userId                 Missing NOT NULL
    "Test" } goes in           → rejected;                    user_id → rejected
  No userId, no role,          invalid role                   before storage
  no type checking             → rejected
```

The document model trades **default enforcement** for **structural flexibility**. Each team decides how far rightward to move on the spectrum — the position is a design choice, not a given.

# Indexes on Documents

Like Topic 5, the document model uses B-tree indexes to avoid collection scans. The same selectivity principle applies.

| Index Type | MongoDB Syntax | Relational Equivalent | When To Use |
|---|---|---|---|
| Single field | `{ "phone": 1 }` | `CREATE INDEX ON users(phone)` | Exact lookups, range queries |
| Compound | `{ "role": 1, "license.expiryDate": 1 }` | `CREATE INDEX ON users(role, expiry)` | Multi-field filters |
| Unique | `{ "phone": 1 }, { unique: true }` | `UNIQUE` constraint | Enforce no-duplicate business keys |
| Multikey | `{ "menu.category": 1 }` | No direct equivalent | Queries on fields inside embedded arrays |
| Text | `{ "businessName": "text" }` | Full-text search extension | Substring search across string fields |

**Multikey indexes** are new — MongoDB indexes each element of an embedded array. No relational equivalent because embedded array data would be separate rows in the relational model.

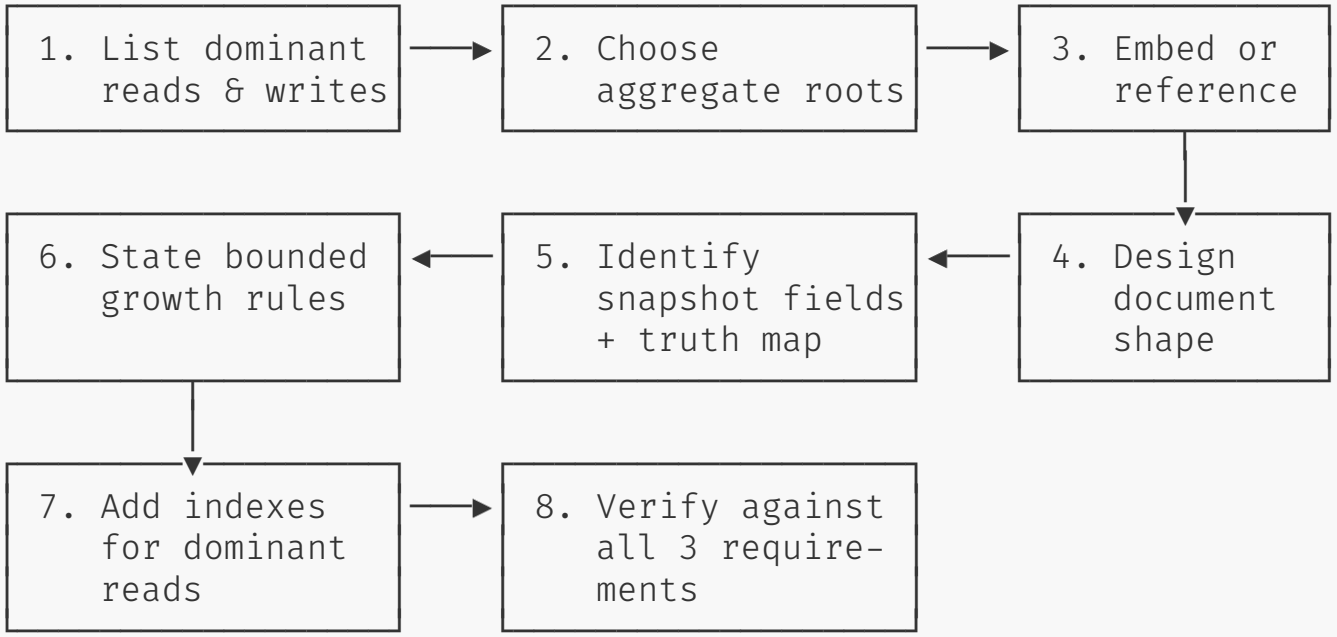
# `explain()` — Before and After Index

```
// Before index — COLLSCAN (full collection
scan):
{
  "queryPlanner": {
    "winningPlan": {
      "stage": "COLLSCAN",
      "filter": { "$and": [ ... ] }
    }
  },
  "executionStats": {
    "totalDocsExamined": 207,
    "totalKeysExamined": 0,
    "executionTimeMillis": 3
  }
}
```

```
// After compound index on { role: 1,
license.expiryDate: 1 } — IXSCAN:
{
  "queryPlanner": {
    "winningPlan": {
      "stage": "IXSCAN",
      "indexName": "role_1_license.expiryDate_1"
    }
  },
  "executionStats": {
    "totalDocsExamined": 17,
    "totalKeysExamined": 17,
    "executionTimeMillis": 0
  }
}
```

The compound index on `(role, license.expiryDate)` lets the engine jump to the "driver" segment of the B-tree, then scan only entries with expiry before 2026-01-01. Same diagnostic pattern as Chapter 5's EXPLAIN: `type: ALL` → `type: ref`.

# The Document Design Loop: Overview




Eight steps, each with a named outcome that feeds the next.

# Steps 1–4: From Workload to Document Shape

| Step | Action | Outcome |
|---|---|---|
| **1. List dominant reads & writes** | "Load driver profile on login", "Update passenger's saved places", "Add merchant menu item" | 3–5 operations the design must make cheap |
| **2. Choose aggregate root(s)** | User profile = natural aggregate root; plans = separate (shared, independent) | One root per collection |
| **3. Embed or reference each entity** | Apply the embed-or-reference test to every related entity | Decision table with justification |
| **4. Design document shape** | Sketch representative JSON for each collection | The JSON the application will read |

Step 4 produces the driver document (Slide 22), the role-specific documents (Slide 24), and the plans collection (Slide 32).

# Embed-or-Reference Decisions: Super-App

| Related entity | Decision | Justification |
|---|---|---|
| Vehicle details (driver) | **Embed** | Retrieved with profile, one per driver, no independent lifecycle |
| Licence info (driver) | **Embed** | Retrieved with profile, bounded (one), updated with driver |
| Saved places (passenger) | **Embed** | Retrieved every session, bounded (≤ 10) |
| Payment methods (all) | **Embed** | Retrieved every session, bounded (≤ 5) |
| Menu items (merchant) | **Embed** | Retrieved with dashboard, bounded (≤ 100) |
| Subscription plan (current) | **Reference** | Shared across thousands, updated independently |
| Subscription snapshot | **Embed** | Historical fact, immutable |
| Ride/order history | **Reference** | Unbounded growth |

# When the Document Model Is the Wrong Choice

The document model optimises for aggregate reads and writes. It's the wrong choice when:

- **The dominant workload requires joins across many independent entities** — analytics spanning orders, products, suppliers, customers. Topic 3's star schema serves this better.
- **Referential integrity must be enforced at the database level** — financial settlement where every payment must reference a valid order. Topic 5's FK constraints are necessary.
- **The data's value lies in relationships between entities** — fraud-detection paths through transaction networks. Topics 8–10 address this.

Applying the document model to the wrong workload recreates the costs it was designed to eliminate, in reverse: instead of expensive joins to assemble data, you get expensive `$lookup` pipelines to connect it.

# Steps 5–8: Discipline and Verification

| Step | Action | Outcome |
|---|---|---|
| **5. Identify snapshots + truth map** | `commissionRateAtSignUp` (snapshot) ↔ `plans.commissionRate` (truth) | Truth map for every duplicated field |
| **6. State bounded-growth rules** | `savedPlaces[]` ≤ 10 (app UI enforces); `menu[]` ≤ 100 (fallback: separate collection) | Bound + rule + fallback per array |
| **7. Add indexes** | `{ "userId": 1 }`, `{ "phone": 1 }` (unique), `{ "role": 1 }`, `{ "menu.category": 1 }` | One index per dominant read |
| **8. Verify against requirements** | R1: shape aligned? R2: no migration? R3: trade-offs named? | Verified design |

If any requirement fails at Step 8, revisit the step that produced the gap.

# Check 1: Hospital Patient Portal (Shape Alignment)

A hospital stores patient records as documents: demographics, allergies (embedded, max ~20), current medications (embedded, max ~15), 5 most recent visits (embedded). Full visit history in a separate `visits` collection (unbounded). The portal API returns the complete patient record on login.

Can the stored document be returned directly as the API response?

> **Yes.** Demographics, allergies, medications, and recent visits are all embedded in the expected shape — the JSON-first design principle holds. Full visit history is deliberately excluded (unbounded growth → referenced) and requires a second query. The aggregate boundary matches the dominant read.

# Check 2: E-Commerce Product Catalogue (Structural Variation)

An e-commerce platform stores products in a single `products` collection: books (`author`, `isbn`), electronics (`specs`, `warrantyYears`), clothing (`size`, `colour`). The business adds a new category: **digital subscriptions** (`billingCycle`, `features[]`, `trialDays`).

Does this require schema migration?

> **No.** A digital subscription is just a new document with new fields in the same collection. Schema-on-read absorbs the variation as data. Cost: if `$jsonSchema` validation uses `category: { enum: [ ... ] }`, the enum must be updated — some migration friction returns for validated fields.

# Check 3: Music Streaming (Bounded Growth Under Pressure)

A music streaming service stores playlists with an embedded `tracks[]` array (bounded at 500). A user imports their personal library: **3,000 tracks** into one "All Songs" playlist.

The bound is exceeded. The fallback activates: move this playlist's tracks to a separate `playlistTracks` collection, referenced by `playlistId`.

> **Predict:** After the fallback, how many `$lookup` stages does the playlist dashboard require for a user with 12 playlists, one of which is the imported library?

# Localised Fallback

> **One `$lookup` —** for the imported library only. The other 11 playlists remain embedded and load in a single read. The fallback applies only to playlists that exceed the bound, not retroactively to all playlists.

The cost of the fallback is **localised** to the playlist that triggered it. The bounded-growth rule is a design trigger, not a permanent constraint — embedded and referenced approaches coexist.

Costs accepted: one `$lookup` per over-bound playlist; increased application complexity to handle both embedded and referenced tracks.

# Check 4: Tax Authority (Snapshot Discipline)

A tax authority stores filings as documents, each embedding `taxBracketAtFiling: "22%"`. The government changes the 22% bracket to 20%.

| Design | Updates Required | Risk |
|---|---|---|
| **Snapshot** (immutable bracket at filing time) | 1 write (update `taxBrackets` collection) | None — filed returns untouched |
| **Current-rate** (bracket references live rate) | 50,001 writes (1 plan + 50,000 filings) | Miss one → incorrect tax calculation |

> Snapshot discipline reduces the update from **50,001 writes to 1 write**. The filed bracket *should* reflect the rate at filing time — the same principle as `commissionRateAtSignUp`.

# Check 5: Pharma Clinical Trials (Analytics Across Boundaries)

Participants embedded in `trials` (bounded ≤ 200/site). Drug suppliers in separate `suppliers` collection (shared, independent).

**Within the aggregate:** "Average participant age per trial site." `$match` → `$unwind` → `$group`. Three stages, one collection.

**Across the boundary:** "Correlation between supplier batch quality and adverse events." Requires `$lookup` to `suppliers` — added complexity and cost.

> **Predict:** Which query is cheaper in the document model? Would the cost difference exist in the relational model?

# The Aggregate Boundary Is Also an Analytics Boundary

> **Within-aggregate** analytics are cheap — data lives in one collection, no `$lookup`. **Cross-aggregate** analytics pay the `$lookup` cost because data is deliberately separated.

In the relational model, both queries use JOINs at roughly uniform cost — the relational model doesn't distinguish "inside" from "outside" an aggregate. This is the flip side of fast aggregate reads: **cheap reads within the boundary, expensive reads across it**.

# Check 6: E-Commerce Orders (Trade-Off Ledger)

Orders as documents with embedded line items. Each line item has `priceAtPurchase` (snapshot) and `productId` (reference).

Walk the trade-off ledger:

| Guarantee | Status | Management strategy |
|---|---|---|
| Referential Integrity | **Lost** — `productId` might reference deleted product | App validates on write; soft-delete products |
| No Redundancy | **Shifted** — `priceAtPurchase` duplicates price | Snapshot is immutable; truth is `products.currentPrice` |
| Access Control | **Lost** — no field-level GRANT | App-layer authorisation filters fields per role |

If any "lost" guarantee has no management strategy, the design fails R3. The ledger applies to *any* document design.

# The Pattern Across All Six

| Question type | Scenarios | Skill tested |
|---|---|---|
| **Design decisions** | 1 (hospital), 2 (e-commerce catalogue) | Apply vocabulary to new domains |
| **Design under pressure** | 3 (music streaming), 4 (tax authority) | Know what to do when assumptions break |
| **Analytical limits** | 5 (pharma), 6 (e-commerce orders) | Honestly name what the document model cannot do cheaply |

The questions are always the same: Is the shape aligned with the dominant read? Are embedded arrays bounded? Are snapshots labelled? Are lost guarantees managed? If you can answer these four for any domain, you can design a document schema for it.

# Course Map

```
Topic 1 (vocabulary, decision framework, four models + validation surveyed)
│
├── Topics 2–5: Relational deep-dive
│   ├── Topic 2: ERD — capturing business rules before tables
│   ├── Topic 3: Normalisation — BCNF, star schema, justified downshifts
│   ├── Topic 4: SQL — FWHOS pipeline, JOINs, aggregation, CTEs
│   └── Topic 5: Enforcement — constraints, transactions, indexes, access control
│
├── Topics 6–10: Alternative models + cross-cutting discipline
│   ├── ▸ Topic 6: Document (MongoDB) — first alternative model;
│   │             schema-on-read, embedding vs referencing
│   ├── Topic 7: Schema validation — XSD and JSON Schema at trust boundaries
│   ├── Topic 8: RDF — knowledge graphs, identifiers, open-world assumption
│   ├── Topic 9: SPARQL — querying knowledge graphs
│   └── Topic 10: Property graph (Neo4j/Cypher) — bounded traversal
│
└── Topic 11: Capstone — multi-model design
```

Topic 6 introduces the trade-off ledger pattern that every subsequent topic will follow.

# Concepts That Recur

| Concept | Introduced Here As | Recurs In |
|---|---|---|
| Aggregate boundary | Document boundary = unit of read/write | Topic 10 (subgraph as traversal unit), Topic 11 (aggregate assignment) |
| Embed vs reference | Nesting vs cross-collection `_id` | Topic 8 (inline literal vs URI reference), Topic 10 (node property vs edge) |
| Snapshot discipline | Immutable copy labelled against truth source | Topic 8 (RDF literal as snapshot), Topic 11 (cross-model strategy) |
| Schema-on-read | Database accepts any shape; app interprets | Topic 8 (RDF open-world assumption), Topic 10 (schema-optional) |
| `$lookup` as LEFT JOIN | Pipeline stage preserving unmatched docs | Topic 9 (SPARQL `OPTIONAL` as LEFT JOIN equivalent) |
| Trade-off ledger | Preserved / lost / shifted from relational baseline | Topics 7–10 (each model's ledger), Topic 11 (combined) |

# Lab Preview

**Topic 6 Lab — Document Modelling: Super-App Personalisation**

- **6 parts**, ~120 minutes total
- **Parts 1–2:** Build two MongoDB designs — Design A (aggregate-first, embedded) and Design B (reference-heavy) — for the same super-app personalisation domain
- **Parts 3–6:** Query, compare, and analyse both designs against the same workload
- **Format:** Predict-first exercises, MongoDB queries, design reasoning

**Tools:** Google Colab with MongoDB (`%%mongodb` magic)

**File:** `06-lab-document-modelling.ipynb`